HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and implemented **PLC programming** logic for **Rockwell/Allen-Bradley** controllers to manage industrial process automation, ensuring seamless operation within cGMP environments.
- Designed and configured **HMI/SCADA systems** using **Siemens and AVEVA (Wonderware)** software for industrial control interfaces, facilitating intuitive user interaction with automated processes.
- Executed field commissioning of control systems, including comprehensive **I/O and instrumentation testing** and full automation system validation, ensuring adherence to GAMP 5 guidelines.
- Authored detailed **engineering design documents**, technical evaluations, and comprehensive test plans for process equipment, aligning with GAMP 5 standards for validated systems.
- Managed and delivered industrial automation projects to customer sites, collaborating with **digital manufacturing/automation engineers** and ensuring project milestones were met.
- Provided technical and automation operational support to Engineering, Manufacturing, and Validation departments, troubleshooting and resolving issues across control systems (**PLC, HMI/SCADA, Communication Networks**).
- Containerized Python-based automation scripts and services using **Docker** and deployed them to an **AWS EC2** instance managed via Jenkins CI/CD pipelines.

SKILLS

PLC Programming – Rockwell/Allen-Bradley, Siemens, AVEVA (Wonderware), Schneider, GAMP 5

HMI/SCADA – Siemens, Rockwell, AVEVA (Wonderware), Schneider

Control Systems & Networks – Profibus, Modbus, DeviceNet, Communication Networks

Engineering & Design – Engineering Design Documents, Technical Evaluations, Test Plans, Functional Design Specifications, I/O Schedules, Panel Wiring Schematics

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins

Scripting & Automation – Python, Shell Scripting

Databases & Data Handling – PostgreSQL, MySQL, Oracle, SQL

PROJECTS

Industrial Automation Control System Design

- Designed and coded automation solutions for industrial processes using **Rockwell PLCs**, integrating with **Siemens HMI/SCADA** for operator interfaces.
- Prepared detailed **I/O schedules and Functional Design Specifications** for a food & beverage manufacturing client, ensuring clarity and compliance with project requirements.
- Conducted on-site **commissioning and fault-finding** for control systems, including instrumentation testing and network diagnostics on **Profibus and Modbus**.

Automated Data Logging and Reporting Pipeline

- Developed a **Python** script to automate the extraction of operational data from industrial control systems via **Modbus**.
- Engineered an automated ETL process using **pandas** to transform and load data into a **PostgreSQL** database, supporting operational analytics.
- Created scheduled reports summarizing key performance indicators, reducing manual reporting effort by approximately **40%** for the operations team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)